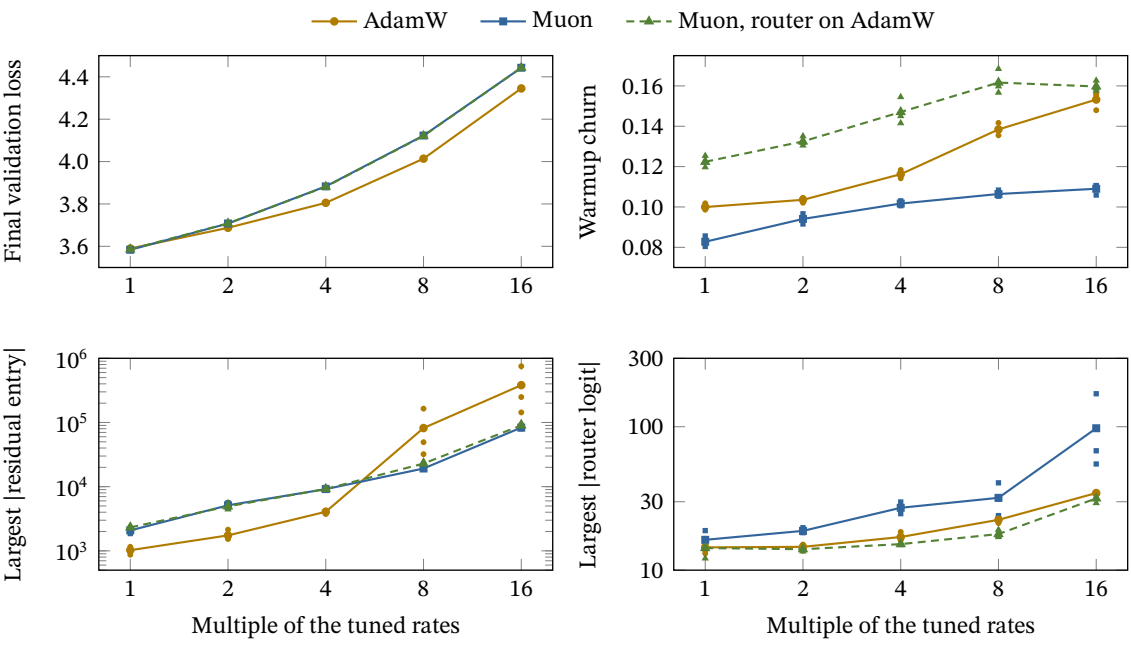

AdamW
Muon
Muon, router on AdamW
Final validation loss
4.4
4.2
4.0
3.8
3.6
Warmup churn
0.16
0.14
0.12
0.10
0.08
Largest |residual entry|
10^6
10^5
10^4
10^3
Largest |router logit|
300
100
30
10
1
2
4
8
16
Multiple of the tuned rates
Multiple of the tuned rates